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A student academic performance prediction model based on the interval belief rule base

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Student performance prediction (SPP) constitutes one of the pivotal tasks in educational data analysis. Outcomes from the prediction enables educators to implement targeted interventions for students. Therefore, developing an effective SPP model is of critical importance. The belief rule base (BRB) is a rule-based modeling approach that integrates expert knowledge and effectively manages uncertain information. Nevertheless, when employing traditional BRB to construct a prediction model, excessive input attributes and reference points may result in a combination explosion. Furthermore, in practical scenarios, the configuration of the model's parameters may be restricted by the limitations of expert knowledge. To overcome these challenges, an SPP model using an interval BRB structure based on the random forest (RF) attribute selection method (IBRB-C) is proposed. The parameters of the IBRB-C model are determined by combining the expert knowledge and the Kmeans++ algorithm. Subsequently, the P-CMA-ES algorithm is applied to optimize the initial model. Ablation experiment is conducted to validate the rationality of the IBRB-C. Finally, case studies on graduate applications and GPA of students demonstrate that the mean squared error (MSE) of the IBRB-C is 0.0024 and 0.1014, respectively. The results of comparative experiments confirm the superiority of the IBRB-C model in predicting student performance.

Focusing on students' academic performance is essential in the educational process. This focus aids in students' personal growth and provides an important framework for educators to evaluate the effectiveness of their teaching methods. The prediction of student performance entails forecasting future academic outcomes by examining past data on student learning[1,2]. Education serves as a crucial foundation for national revitalization and societal advancement, and SPP holds considerable importance[3–6]. Educators are accessible to offer individualized, focused assistance and prompt intervention for students[7,8]. As society places greater importance on fostering an ideal educational environment, the SPP holds significant promise for steering and overseeing the teaching process[9]. Therefore, an efficient SPP model is crucial to stay aligned with the swift advancements in educational information technology[10,11].

Currently, researchers in related fields have proposed a series of student performance prediction methods. These methods can be broadly classified into three categories: data-driven, physical-driven, and hybrid-driven models[12]. Data-driven models analyze data samples to identify patterns or correlations. Mi proposed a model to predict student performance based on the neural network of the radial basis function (RBF) neural network[13]. The model accurately predicted performance by analyzing the learning situation of students, demonstrating the potential of machine learning in the field of education. In recent years, with significant progress in deep learning (DL), SPP methods based on DL have made further breakthroughs[14]. Sun used the adaptive federated deep learning algorithm to predict college students' sports performance, studied the influencing factors, and presented improvement suggestions[15]. Dien et al. combined deep learning, data transformation, and factor analysis methods. They proposed several data pre-processing techniques to predict student performance combined with models[16]. However,It should be noted that the training of data-driven models relies on a large number of data samples. In addiction, prediction results lack interpretability because they are black-box models.

Physical-driven models are developed upon theoretical foundations and empirical guidelines. These models rely on a comprehensive understanding and mathematical depiction of the system's physical process. Ouyang et al.[17] used an artificial intelligence prediction model combined with the learning analysis method to improve the effect of students in a collaborative learning environment. Based on self-regulation and prediction of learning analysis, Cogliano et al. monitored students with difficulties learning biology and provided support[18]. Physical-driven models have some advantages due to their theoretical integrity and interpretability. However, it can be

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difficult for these models to understand the interaction of various influencing factors and to deal with complex problems.

Hybrid-driven models integrate the advantages of data-driven and physical-driven models. Hao et al.[19] used Bayesian networks to predict student MOOC performance and suggested improvement measures. Chiu et al.[20] used a hidden Markov model to predict student performance. Raes et al.[21] proposed the concept of a blended learning environment to combine elements of on-site and distance learning to adapt to educational needs during the COVID-19 pandemic. These hybrid-driven SPP models combine the experience of educational experts with sample data from relevant cases. Therefore, their results have great advantages in accuracy and interpretability. Educators are encouraged to prioritize the student-centered principles. When developing the prediction model, it is essential to comprehensively utilize the expertise accumulated by educational professionals. Nevertheless, pertinent data in the educational domain frequently exhibit a degree of subjectivity and uncertainty, thereby influencing the outcomes of the model[22]. Therefore, prediction models must be able to handle uncertain information.

In this study, belief rule base (BRB) is selected to build the model. BRB is a hybrid-driven model proposed by Yang et al.[23], which is based on IF-THEN rules and evidential reasoning (ER)[24]. BRB effectively solves uncertainty and ambiguity in the data collection process by using the belief distribution to describe qualitative or quantitative data[25,26]. Regarding education issues, BRB has good interpretability, which enables educators to understand the process of achieving the results. Liang et al.[27] constructed the HBRB-I model to predict student performance. Gong et al.[28] also proposed a hierarchical BRB model to predict global educational inequality. Liu et al.[29] used data mining technology to analyze educational data, and adjusted the balance between precision and complexity of the prediction model with AIC. In addition to interpretability, researchers are increasingly focusing on the robustness of BRB models[30,31]. Therefore, BRB has demonstrated increasingly broader application prospects in fault detection and health assessment of complex systems[32–34]. Yin et al. further considered the attribute reliability and reverse causal inference based on interpretable BRB modeling, successfully applied to the health state assessment of aeronautical relay and diesel engines[35,36]. Lian et al. proposed the linguistic Z-Number belief rule base considering the reliability of knowledge, which demonstrated excellent performance while retaining a high transparency in modeling process[37–39].

However, the combination explosion will be caused by too many input attributes or reference points in traditional BRB models. Rule reduction and hierarchical structure are used to reduce the complexity of the BRB. Zhou et al. conducted a rule evaluation using a statistical utility threshold[40,41]. Kalia et al. introduced dimensionality reduction methods such as multidimensional scaling (MDS), principal component analysis (PCA), Isomap, and gray objective (GT)[42–45]. Zhang et al.[46] used the BRB model to predict tunnel squeezing and avoided the combination explosion by building the hierarchical structure. However, rule reduction methods will lose model accuracy and have limitations in filtering important attributes. The initial hierarchical BRB model is unable to train the intermediate layers, increasing the uncertainty of the model. Therefore, Sun et al. proposed an interval BRB model (IBRB), which can better cope with uncertainty and diversity by dividing input attributes into reference intervals[47]. In the field of education, student performance often presents a normal distribution. IBRB model can make full use of data samples, effectively integrate multidimensional data, and improve the accuracy of the prediction results.

The following two problems need to be solved when construct the SPP model based on IBRB. (1) A series of factors affect students' academic performance. The question of how to select the appropriate input attributes must be considered. (2) Setting the reference intervals is vital for model construction, and the inefficient expert knowledge will affect the process. To solve these problems, in this paper, an interval BRB student performance prediction model based on random forest (RF) attribute selection is proposed. First, features with the highest importance are chosen as input attributes. Expert knowledge is combined with the Kmeans++ algorithm to divide reference intervals. Then, ER rule is used as the inference mechanism. Finally, optimization of the initial parameters is performed by the P-CMA-ES algorithm to improve the accuracy.

The main contributions of this paper are as follows.

(1) This paper proposes the IBRB-C student performance prediction model.
(2) An interval BRB structure based on RF attribute selection is constructed to solve the combination explosion, and the reliability of each belief rule is considered in the model inference process.
(3) Expert knowledge and the Kmeans++ algorithm are combined to determine the reference intervals of the model. This method can compensate for the lack of expert knowledge and improve the accuracy of the IBRB-C model.

The main structure of this paper is as follows. In section "Problem formulation", three problems are elaborated to build the IBRB-C model. Section "IBRB-C student performance prediction model" describes the construction, inference, and optimization process of the IBRB-C model. In section "Case study", two experimental cases are combined to verify the effectiveness of the IBRB-C model. Section "Conclusions" concludes the paper and discusses future work.

Problem formulation

Problem 1: When designing the structure of IBRB-C model, it is indispensable to determine the input attributes and reference intervals. Therefore, the first question is how to choose input attributes. The process can be expressed as follows.

$$\{x_1, x_2, \ldots, x_M\} = f\left(X_1, X_2, \ldots, X_K\right)(M < K) \tag{1}$$

where $X_1, X_2, \ldots, X_K$ denote all available attributes. $f(\cdot)$ denotes the attribute selection function. $x_1, x_2, \ldots, x_M$ denote the selected input attributes of the model.

The setting of the reference intervals is generally based on expert knowledge. However, if expert knowledge is insufficient, it is difficult to ensure the precision and integrity of the model. Therefore, a reference interval determination method combined with expert knowledge is needed. The process can be described as follows.

$$\left\{r_i^1, r_i^2, \ldots, r_i^m\right\} = E\left(D\left(X_i\right)\right), i = \{1, 2, \ldots, M\} \tag{2}$$

where $\left\{r_i^1, r_i^2, \ldots, r_i^m\right\}$ and X_i denote the set of reference points to be determined and the data samples of the i_{th} input attribute, respectively. $D(\cdot)$ denotes the determination algorithm. $E(\cdot)$ denotes the adjustments made by experts.

Problem 2: How to properly design the inference process of the model. The inference process of IBRB is different from that of traditional BRB. The process can be described as follows.

$$u = g(x \cdot \alpha) \tag{3}$$

where *x* represents the evaluation levels of the model. α represents the set of parameters in the inference process. *u* represents the prediction result. $g(\cdot)$ represents the inference function.

Problem 3: How to design the optimization process of the model. As a hybrid-driven model, the initial model of BRB needs to be optimized. The process can be described as follows.

$$\alpha_{\text{ best}} = \text{optimize}(\theta, \beta, \xi) \tag{4}$$

where θ and β represent inputs and outputs. ε represents the set of parameters used in the optimization process. $\alpha_{\text{ best}}$ represents the set of optimal model parameters. $optimize(\cdot)$ represents the optimization algorithm.

IBRB-C student performance prediction model

This section outlines the construction methodology of the IBRB-C model. The IBRB model substitutes the reference points of the input attributes with intervals. Upon encountering a data sample within a reference interval, the corresponding belief rule is activated. These activated rules are involved in the inference and optimization process of the model. Typically, the distribution of student scores is either scattered or displays a segmental concentration. By employing intervals to depict the data, the IBRB-C model more effectively captures the uncertainty information, making it more appropriate for practical application.

This section is structured as follows. Subsection "Construction process of the IBRB-C model" describes the construction process of the IBRB-C model. Subsection "Inference process of the IBRB-C model" explains the inference procedure of the model. Subsection "Optimization process of the IBRB-C model" presents the optimization procedure of the IBRB-C model.

Construction process of the IBRB-C model

The IBRB-C is modeled based on IF-THEN rules, which consist of a series of belief rules. Assuming that the input attributes are independent of each other, the relationship between the evaluation levels and the result of the model can be described as follows.

$$\begin{aligned}
&Belief Rule_c: \\
&if\ x_1 \in [a_1, b_1] \vee x_2 \in [a_2, b_2] \vee \ldots \vee x_M \in [a_M, b_M] \\
&Then\ result\ is\ \left((\mathrm{O}_1, \beta_{1,c}), (\mathrm{O}_2, \beta_{2,c}), \ldots, (\mathrm{O}_N, \beta_{N,c})\right) \\
&with\ rule\ weight\ \omega_c \\
&and\ rule\ reliability\ r_c \\
&\sum_{i=1}^{N} \beta_{i,c} \leq 1, c \in \{1, 2, \ldots, L\}
\end{aligned} \tag{5}$$

where L , M and N denote the number of belief rules, input attributes, and evaluation levels of the model, respectively. x_i denotes the input attributes. $[a_i, b_i]$ denotes the reference intervals. O_i denotes the evaluation levels. $\beta_{i,c}$ denotes the belief degree of each evaluation level under the c_{th} belief rule. ω_c and r_c denote the weight and reliability of the c_{th} belief rule, respectively.

It is worth noting that, unlike the traditional BRB, the input attributes of the proposed model are connected by the disjunctive notation. The main reason is that interval BRB uses interval addition to combine rules instead of Cartesian product. In this way, for the same number of input attributes and reference points, the number of rules in interval BRB models will be greatly reduced[26].

The general construction process of the IBRB-C model is shown in Fig. 1.

Attribute selection of the IBRB-C model

It is crucial to acknowledge that the integration way of rules in the IBRB model enables more input attributes compared to the traditional BRB[26]. However, in practical scenarios, the factors affecting student academic performance are multifaceted. Additionally, the gathered data samples often encompass various types. For instance, some features might include binary data, where 0 denotes "no" and 1 denotes "yes". The selection

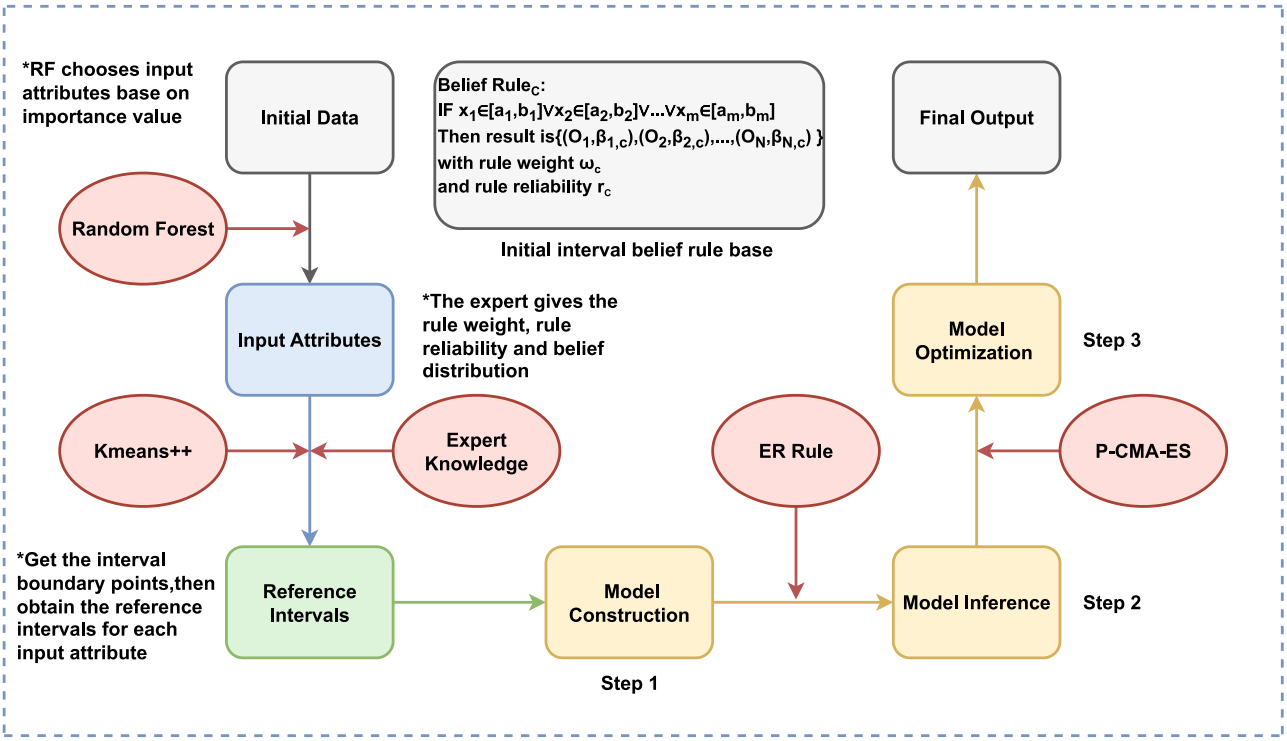


Fig. 1. The construction process of the IBRB-C model.

of these features as input attributes is problematic. They complicate the definition of reference intervals and pose difficulties in assessing their significance. Meanwhile, the process of manually selecting features is easily influenced by subjectivity, which affects the persuasiveness and rationality of the modeling process. Therefore, this study integrates an attribute selection module within the IBRB-C model to achieve a balance between prediction accuracy and model simplicity.

The main difficulty in attribute selection is assessing the significance of each feature. To address this, this study employs the random forest (RF) approach. RF is an ensemble learning method that develops multiple decision trees and combines their output to generate the final prediction. Consider that the input feature matrix is $X \in \mathbb{R}^{n \times p}$ and the target attribute is $y \in \mathbb{R}^n$, where n and p denote the number of samples and features, respectively. The importance value for each feature can be calculated as follows.

Step 1: Construct the model. The random forest comprises T regression trees, where each tree $t_i (i = 1, \ldots, T)$ fits the data using the least squares method. The predicted value based on the input attribute x is denoted as $\hat{y}_t(x)$. The final prediction can be calculated as follows.

$$\hat{y}(x) = \frac{1}{T} \sum_{t=1}^{T} \hat{y}_t(x) \tag{6}$$

Step 2: Splitting. In each tree, for a given feature X_j and a split point s, the objective of the split process is to minimize the loss function. The loss function can be formulated as follows.

$$L(j,s) = \sum_{i \in X_L} (y_i - \bar{y}_{left})^2 + \sum_{i \in X_R} (y_i - \bar{y}_{right})^2 \tag{7}$$

where X_L and X_R denote the sets of left and right child nodes after splitting, respectively. $\bar{y}_{left}$ and $\bar{y}_{right}$ denote the mean values of the target attribute of the left and right child nodes after splitting, respectively.The split point s is a sample of the feature X_j.

Step 3: Calculate the importance value of the feature. The value is determined on the contribution of each feature to the target attribute y during the split process of each tree in the model. It can be calculated as follows. First, for each tree $t_i (i = 1, \ldots, T)$,the total loss of feature X_j across all node splits is calculated as follows.

$$\text{Importance}_{t}(X_j) = \sum_{s(X_j)} \Delta L(j,s) \tag{8}$$

$$\Delta L(j,s) = L_{\text{before}} - L_{\text{after}} \tag{9}$$

where $S(X_j)$ denotes all the splits containing the feature X_j. $\Delta L(j,s)$ denotes the change in the splitting loss. L_{before} and L_{after} denote the amount of loss before and after splitting,respectively. Then, the importance value of feature X_j can be obtained by averaging over all T trees.

$$\text{Importance}(X_j) = \frac{1}{T}\sum_{t=1}^{T} \text{Importance}_{t}(X_j) \tag{10}$$

Finally, the importance values of all features can be normalized as follow.

$$\text{Normalized Importance}(X_j) = \frac{\text{Importance}(X_j)}{\sum_{k=1}^{p} \text{Importance}(X_k)} \tag{11}$$

Step 4: Two features with the highest values are selected as input attributes of the IBRB-C model.

Determination of the reference intervals of the IBRB-C model

Reference points for the BRB are generally established using expert knowledge. However, in practical scenarios, such insights may prove insufficient. Zhang et al. have proposed a Kmeans++ algorithm with error constraints to generate reference points, which has been utilized to forecast the behavior of complex systems[48]. Building upon this algorithm, Liu et al. implemented it within a BRB-based SPP model[29]. Therefore, reference intervals for the IBRB-C model are determined by combining expert knowledge with the Kmeans++ algorithm in this study.

It is important to note that the point set generated by the Kmeans++ algorithm excludes endpoints. The lack of endpoints may undermine the comprehensiveness of the BRB model[49]. To remedy this deficiency, the Kmeans++ algorithm is initially employed to establish the set of interval boundary points, followed by adding endpoints with expert knowledge. The procedures are outlined as follows.

Step 1: Select the initial center. C cluster centers are preset based on the reality of the research. A sample point of the input attribute X_1 is randomly selected as the first cluster center c_1.

Step 2: Calculate the distance. The distance $D(x_i)$ between each sample point x_i and the cluster center c_j can be calculated as follows.

$$D(x_i) = \min_{c_j \in C} \|x_i - c_j\|^2 \tag{12}$$

where C denotes the cluster centers that have been selected. $\|x_i - c_j\|$ denotes the Euclidean distance between x_i and c_j.

Step 3: Select the next cluster center. Each sample x_i is chosen as the next cluster center with probability proportional to $D(x_i)$. Randomly selected according to the following probability distribution.

$$P(x_i) = \frac{D(x_i)}{\sum_{x_r \in X} D(x_r)} \tag{13}$$

where X denotes the samples of this attribute. x_r denotes the remaining samples except x_i.

Step 4: Repeat the above steps until all cluster centers are selected.

Step 5: Assign. For each sample x_i, it's assigned to the nearest cluster center c_j, and the process can be described as follows.

$$\text{cluster}(x_i) = \arg\min_{c_j} \|x_i - c_j\|^2 \tag{14}$$

where $\text{cluster}(x_i)$ denotes the cluster center that x_i should be assigned.

Step 6: Update. For each cluster center, its position is recalculated as the mean of all samples that belong to it. The formulation is described as follows.

$$c_j = \frac{1}{|s_j|} \sum_{x_i \in S_j} x_i \tag{15}$$

where S_j denotes the set of samples belonging to the cluster center c_j.

Step 7: Repeat Step 5 and 6 until all interval boundary points are established. Now, the initial set of interval boundary points is obtained. The endpoints are added with expert knowledge to ensure meaningfulness. For example, the Kmeans++ algorithm provides the initial points a_1, a_2, a_3, and the endpoints selected by experts are b_1, b_2. Therefore, the final interval boundary points are b_1, a_1, a_2, a_3, b_2. The reference intervals can then be set as $[b_1, a_1]$, $[a_1, a_2]$, $[a_2, a_3]$and $[a_3, b_2]$.The process is repeated until each input attribute has suitable reference intervals.

Inference process of the IBRB-C model

The inference process of the traditional BRB uses ER parsing algorithm, without considering the reliability of each belief rule[49]. Cheng et al. designed a new inference algorithm for the IBRB, which improved the performance of the model[26]. The inference process of the IBRB-C model can be summarized as follows. After judging the activation rules based on the input information, the ER rule is used to fuse the belief rules. Finally, the expected utility is obtained by utility calculation as the model prediction results. The process can be described as follows.

Step 1: Assume that the rule activated by the IBRB-C model is independent evidence and there are C independent evidence $e_c(c = 1, \ldots, L)$. The belief distribution of the evidence can be obtained from the activated rule. The c_{th} evidence is represented as follows.

$$e_c = \{(\theta_n, \beta_{n,c}), (\Theta, \beta_{\Theta,c})\}, n = 1, \ldots, N \tag{16}$$

where $\theta_{\mathrm{n}}(n = 1, \ldots, N)$ denotes the evaluation level. $\beta_{n,c}$ denotes the belief degree that is evaluated as level θ_n under the evidence e_c. $\Theta = \{\theta_1, \ldots, \theta_n\}$ denotes the discernment frame. $\beta_{\Theta,c}$ denotes the global ignorance.

Step 2: The rule weight ω_c and the rule reliability r_c of independent evidence e_c are given by the expert in advance. The weighted belief distribution of the c_{th} evidence with reliability can be obtained as follows.

$$m_c = \left\{(\theta_n, \tilde{m}_{\theta,c}), \left(\beta(\Theta), \tilde{m}_{\beta(\Theta),c}\right)\right\}, \forall \theta \subseteq \Theta \tag{17}$$

where $\beta(\Theta)$ denotes the power set. $\tilde{m}_{\theta,c}$ denotes the mixed probability quality with indicator c under the level θ_n. $\tilde{m}_{\theta,c}$ satisfies:

$$\tilde{m}_{\theta,c} = \begin{cases} 0, & \theta = \varnothing \\ C_{rw,c} m_{\theta,c}, & \theta \subseteq \Theta, \theta \neq \varnothing \\ C_{rw,c}(1 - r_c), & \theta = \beta(\Theta) \end{cases} \tag{18}$$

$$\sum_{\theta \in \Theta} \tilde{m}_{\theta,c} + \tilde{m}_{\beta(\Theta),c} = 1 \tag{19}$$

$$m_{\theta,c} = \omega_c \beta_{\theta,c} \tag{20}$$

where $C_{rw,c} = 1/(1 + \omega_c - r_c)$ denotes the regularization coefficient. $m_{\theta,c}$ denotes the basic probability quality of indicator c under the level θ_n.

Step 3: C independent evidences $e_c(c = 1, \ldots, L)$ for proposition θ can be obtained by iterating the following equations.

$$\widehat{m}_{e,(b)} = [(1 - r_b) m_{\theta,e(b-1)} + m_{p(\Theta),e(b-1)} m_{\theta,b}] + \textstyle\sum_{A \cap B = \theta} M_{A,e(b-1)} m_{B,b}, \forall \theta \in \Theta \tag{21}$$

$$\widehat{m}_{P(\Theta),e(b)} = (1 - r_b) m_{\beta(\Theta),e(b-1)} \tag{22}$$

$$m_{n,e(b)} = \begin{cases} 0, & \theta = \varnothing \\ \dfrac{\widehat{m}_{n,e(b)}}{\sum_{A \subseteq \Theta} \widehat{m}_{A,e(b)} + \widehat{m}_{\beta(\theta),e(b)}}, & \theta \subseteq \Theta, \theta \neq \varnothing \end{cases} \tag{23}$$

$$\beta_{n,e(b)} = \begin{cases} 0, & \theta = \varnothing \\ \dfrac{\widehat{m}_{n,e(b)}}{\sum_{A \subseteq \Theta} \widehat{m}_{A,e(B)}} & \theta \subseteq \Theta, \theta \neq \varnothing \end{cases} \tag{24}$$

where $m_{\theta,e(1)} = m_{\theta,1}$,$m_{\beta(\Theta),e(1)} = m_{\beta(\Theta),1}$. $m_{\theta,e(b)}$ denotes the joint probability density of the b pieces of evidence. $\beta_{\theta,e(b)}$ denotes the joint belief degree of the b pieces of evidence.

Step 4: Use $u(\theta_n)$ denotes the utility of the evaluation level θ_n. The expected utility can be calculated as follows.

$$u = \sum_{n=1}^{N} u(\theta_n)\, \beta_{\theta,e(b)} \tag{25}$$

where u is the result of Eq. (3). The general flow chart of the ER rule is shown in Fig. 2.

Optimization process of the IBRB-C model

When the BRB model demonstrates insufficient accuracy, this may result from rules that are not properly allocated within the belief distribution[50]. Therefore, the IBRB-C model focuses on optimizing the initial parameters to minimize the MSE.

Various algorithms have been employed in the optimization of models. A widely adopted optimization algorithm for BRB is the P-CMA-ES algorithm[51]. The P-CMA-ES algorithm is a parallel optimization method based on an adaptive evolution strategy of the projected covariance matrix. It mimics the principles of biological evolution and demonstrates strong robustness and rapid convergence. Therefore, the P-CMA-ES algorithm is utilized to optimize the IBRB-C model.

The objective function of the optimization algorithm can be formulated as follows.

$$\alpha_{\text{ best }} = \text{optimize}\left(\theta_n, \beta_{n,c}, \xi_c\right) = \min MSE\left(\theta_n, \beta_{n,c}, \xi_c\right) \tag{26}$$

where $MSE\left(\theta_n, \beta_{n,c}, \xi_c\right)$ represents the prediction accuracy of the model, which is calculated as follows.

$$MSE\left(\theta_n, \beta_{n,c}, \xi_c\right) = \frac{1}{P} \sum_{p=1}^{P} \left(\text{output}_{\text{ predict }} - \text{output}_{\text{ actual}} \right)^2 \tag{27}$$

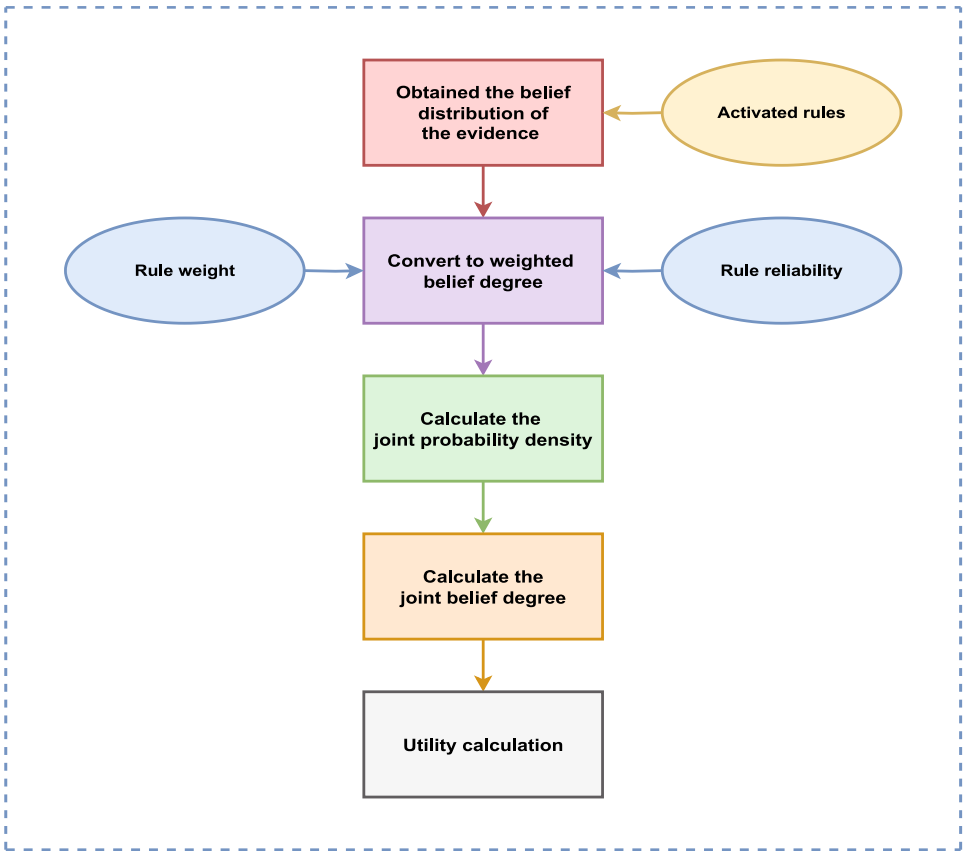


Fig. 2. The flow chart of ER rule.

where P represents the number of samples. $\text{output}_{\text{predict}}$ represents the prediction result of the model. $\text{output}_{\text{actual}}$ represents the actual value. The prediction result can be obtained as follows.

$$\text{output}_{\text{predict}} = \sum_{n=1}^{N} u(\theta_n)\,\beta_{\theta,e(b)} \tag{28}$$

The constraints for parameter optimization are given as follows.The optimization process of the model is shown in Fig. 3.

$$\begin{gathered} \min MSE(\theta_n, \beta_{n,c}, \xi_c) \\ \text{s.t.} \\ \sum_{n=1}^{N} \beta_{n,c} = 1 \quad n \in \{1,\ldots,N\}, c \in \{1,\ldots,L\} \\ 0 \le \theta_n \le 1 \quad n \in \{1,2,\ldots,N\} \\ 0 \le \xi_c \le 1 \quad c \in \{1,2,\ldots,L\} \\ 0 \le \beta_{n,c} \le 1 \quad n \in \{1,\ldots,N\}, c \in \{1,\ldots,L\} \end{gathered} \tag{29}$$

The steps to execute the P-CMA-ES algorithm can be summarized as follows.

Step 1: Parameter initialization.

$$\Omega^0 = \{\theta_1,\ldots,\theta_L,\beta_{1,1},\ldots,\beta_{N,L},\xi_1,\ldots,\xi_L\} \tag{30}$$

where Ω^0 represents the initial parameters to be optimized.

Step 2: Obtain each generation through a sampling operation.

$$\Omega_i^{g+1} \sim w^g + \varepsilon^g N(0, C^g), i = 1,\ldots,\lambda \tag{31}$$

where Ω_i^{g+1} denotes the i_{th} solution in the $(g+1)_{th}$ generation. w^g and ε^g denote the generation strength and step size in g_{th} generation, respectively. C^g denotes the covariance matrix in the g_{th} generation. $N(*)$ and λ denote the normal distribution and the number of descendants, respectively.

Step 3: Project the solution onto the feasible hyperplane to satisfy the given constraints. This hyperplane can be represented as follows.

$$\begin{gathered} \Omega_i^{g+1}(1+n_e\times(j-1):n_e\times j) = \Omega_i^{g+1}(1+n_e\times(j-1):n_e\times j) - \\ A_e^T \times \left(A_e \times A_e^T\right)^{-1} \times \Omega_i^{g+1}(1+n_e\times(j-1):n_e\times j) \times A_e \end{gathered} \tag{32}$$

$$A_e \Omega_i^g (1+n_e\times(j-1):n_e\times j) = 1 \tag{33}$$

where A_e denotes the parameter vector.In the solution Ω_i^g , n_e and j denote the number of constraint variables and the number of equality constraints, respectively.

Step 4: Update the mean of the next generation.

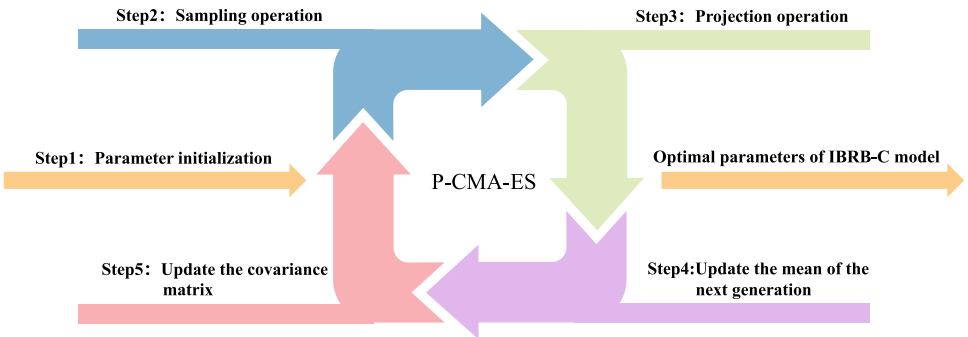


Fig. 3. Optimization process of P-CMA-ES algorithm.

$$w^{g+1} = \sum_{i=1}^{\tau} h_i \Omega_{i:\lambda}^{g+1} \tag{34}$$

where h_i denotes the weight coefficient. $\Omega_{i:\lambda}^{g+1}$ denotes the i_{th} solution of the λ solution in $(g+1)_{th}$ generation. τ denotes the population size of the offspring.

Step 5: Update the covariance matrix.

$$\begin{aligned} C^{g+1} &= (1 - c_1 - c_2)\, C^g + c_1 P_c^{g+1} \left(P_c^{g+1}\right)^T + \\ & c_2 \sum_{i=1}^{v} h_i \left(\frac{K_{i:\lambda}^{g+1} - \varphi^g}{\rho^g}\right) \left(\frac{K_{i:\lambda}^{g+1} - \varphi^g}{\rho^g}\right)^T \end{aligned} \tag{35}$$

where ρ^g denotes the growth step of the g_{th} generation. c_1 and c_2 denote the learning rate. P_c^{g+1} denotes the evolution path of the $(g+1)_{th}$ generation. φ^g denotes the number of variables in descending order in the g_{th} generation. $K_{i,\lambda}^{g+1}$ denotes the i_{th} parameter vector λ of the $(g+1)_{th}$ generation.

Step 6: Recursively perform steps 1 to step 5 until the best parameters are obtained.

Case study

In this section, two case studies were used to verify the validity and superiority of the IBRB-C student performance prediction model. In Subsection "IBRB-C model for prediction of graduate admission success rate", a graduate application dataset containing 500 data sets was researched for the ablation and comparison experiments. In Subsection "IBRB-C model for prediction of students'GPA", a dataset containing nearly 2,400 data sets was studied to predict students' GPA, verifying the prediction performance of IBRB-C model on the large-scale dataset.

IBRB-C model for prediction of graduate admission success rate

Backgrounds of the dataset

Data collected for this case study relates to a group of Indian students. The dataset can be publicly obtained from https://www.kaggle.com/datasets/mukeshmanral/graduates-admission-prediction (accessed on 12 July 2022), which has 500 sets of data. This dataset consists of one target attribute and seven input features, including test scores, written materials, and learning experience. The information are detailed in Table1.

Construct IBRB-C student performance prediction model

The initial IBRB-C model needs to build a belief rule base, which is constructed as follows:

$$\begin{aligned} & Belief Rule_c : \\ & if\ x_1 \in [a_1, b_1] \vee x_2 \in [a_2, b_2] \\ & Then\ result\ is\ ((\mathrm{O}_1, \beta_{1,c}), (\mathrm{O}_2, \beta_{2,c}), (\mathrm{O}_3, \beta_{3,c}), (\mathrm{O}_4, \beta_{4,c})) \\ & with\ rule\ weight\ \omega_c \\ & and\ rule\ reliability\ r_c \\ & \sum_{i=1}^{N} \beta_{i,c} \le 1, c \in \{1, 2, \ldots, L\} \end{aligned} \tag{36}$$

where x_1 and x_2 represent the input attributes. $O_i (i = 1, 2, 3, 4)$ represents the evaluation level as the prediction result of the model. In this case study, the evaluation levels are low (O_1), medium (O_2), high (O_3), and very high (O_4).

Index	Feature	Data type	Type
1	GRE	Integer	Input
2	TOEFL	Integer	Input
3	University ranking	Integer	Input
4	Statement of purpose	Float	Input
5	Letter of recommendation	Float	Input
6	CGPA	Float	Input
7	Research	Bool	Input
8	Chance of admit	Float	Target

Table 1. The detailed information of the dataset.

Index	Feature	Importance value
1	GRE	0.265
2	TOEFL	0.212
3	University ranking	0.063
4	Statement of purpose	0.032
5	Letter of recommendation	0.021
6	CGPA	0.402
7	Research	0.005

Table 2. The importance value of each feature.

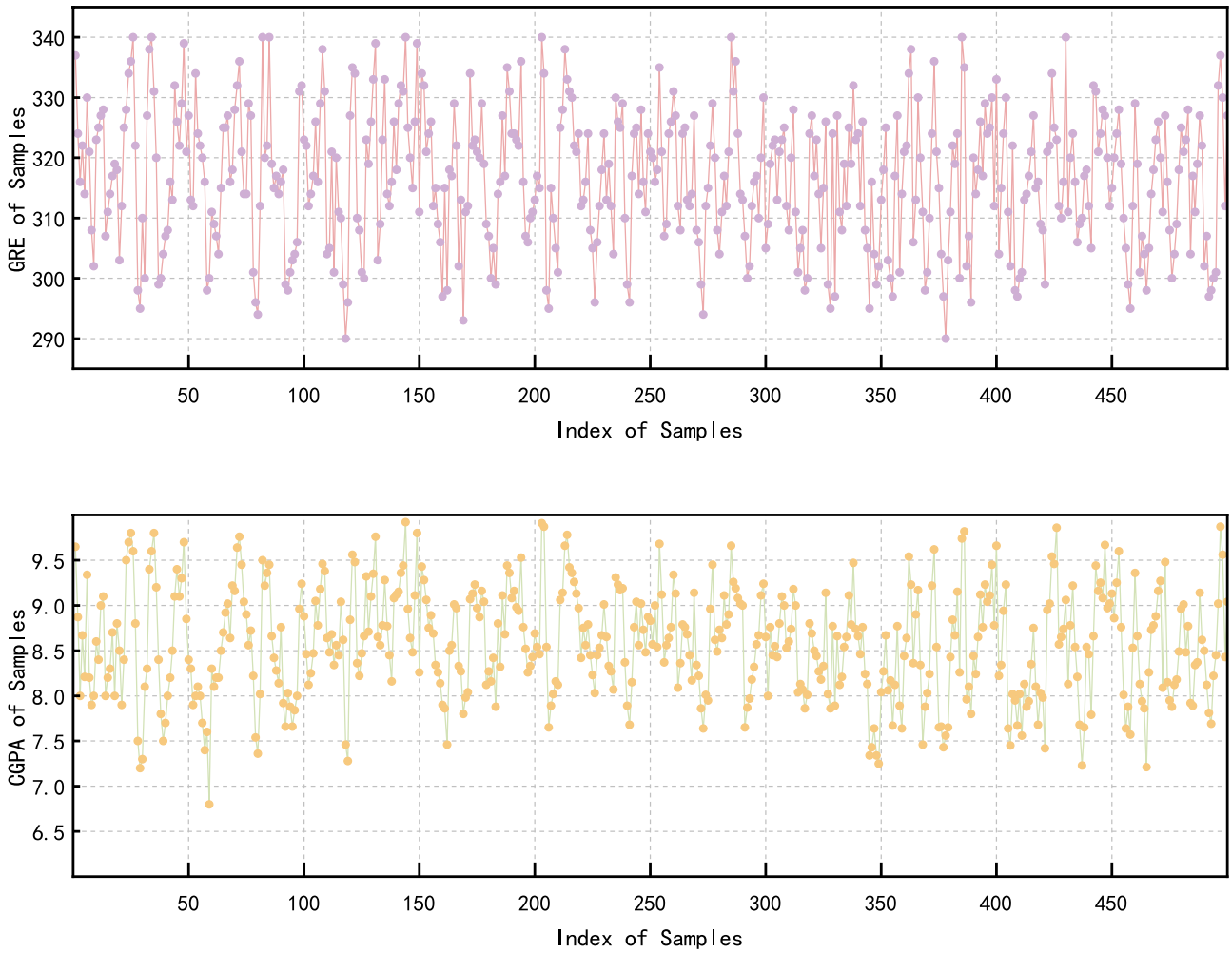


Fig. 4. Data distribution of GRE and CGPA.

Random forest method was used to select the input attributes of the IBRB-C model. The importance values of each feature are shown in Table 2. Two features with the highest values, GRE and CGPA, were taken as input attributes of the IBRB-C model. Though TOEFL has a comparable importance value to the GRE, its data distribution is too concentrated to obtain enough cluster centers by Kmeans++ algorithm, so it has not been considered as a input attribute. The data distribution of GRE and CGPA are shown in Fig. 4. In this case study, 100 data are randomly selected for testing, and the remaining data are used for training. The experimental environment includes Windows 11 and a 9_{th}Gen Intel (R) Core (TM) i7-9750H processor. The experiment is carried out using Matlab R2024a software.

The upper and lower bounds of the attributes are presented in Table 3. When conducting the Kmeans++ algorithm, the number of cluster centers C needs to be set in advance. By analyzing the data distribution, 18 reference intervals are decided for each attribute, which means 19 interval boundary points. Since the results of algorithm exclude endpoints, the value of C in this case study is 19-2 = 17. As illustrated in Table 4, 36 reference

Index	Feature	Upper bound	Lower bound	Type
1	x_1(GRE)	340	290	Input
2	x_2(CGPA)	9.92	6.80	Input
3	Chance of admit	0.97	0.34	Target

Table 3. The details of input and target attributes.

Index	GRE	CGPA
1	(290,296)	(6.80,7.28)
2	(296,299)	(7.28,7.48)
3	(299,302)	(7.48,7.66)
4	(302,305)	(7.66,7.87)
5	(305,308)	(7.87,8.01)
6	(308,311)	(8.01,8.16)
7	(311,312)	(8.16,8.30)
8	(312,315)	(8.30,8.48)
9	(315,317)	(8.48,8.65)
10	(317,320)	(8.65,8.77)
11	(320,322)	(8.77,8.88)
12	(322,324)	(8.88,9.01)
13	(324,327)	(9.01,9.13)
14	(327,329)	(9.13,9.25)
15	(329,332)	(9.25,9.42)
16	(332,335)	(9.42,9.62)
17	(335,339)	(9.62,9.82)
18	(339,340)	(9.82,9.92)

Table 4. The reference intervals of the IBRB-C model.

Group	Reference values	Data number within intervals
1	(0.35,0.65,1)	144,356
2	(0.35,0.55,0.75,1)	61,221,218
3	(0.35,0.5,0.65,0.8,1)	35,109,199,155

Table 5. Three different sets of reference values.

intervals are composed of points determined jointly by the Kmeans++ algorithm and expert knowledge. Each rule in IBRB inherently represent a belief distribution and can serve directly as evidence[26], which means 36 belief rules are consequently generated.

From Table 3, the value of predictive target is between 0.34 and 0.97, so boundary of the reference values can be set as 0.35 and 1, with a difference of 0.65. When determining the remaining values, the difference between adjacent ones should be similar. Table 5 presents three sets of reference values, with an approximate difference of 0.3, 0.2, and 0.15, respectively. Compared to the second group, data distribution of other groups are relatively more uneven. In addition, the number of values in the second group is also appropriate. Therefore, (0.35, 0.55, 0.75, 1) was chosen as the reference values set.

In this case study, the initial rule weight and the rule reliability were initialized to 1. If the inputs of attributes x_1 and x_2 are 295 and 7, the two activated rules correspond to the intervals (290, 296) and (6.8, 7.28). These two rules are fused using the ER rule, followed by utility calculation to yield the final evaluation level. In the IBRB model, although the expanded range of reference points mitigates the uncertainty of expert knowledge, the belief degree, rule weight, and rule reliability still rely on it. Therefore, the IBRB-C model requires an optimization algorithm to refine the parameters. The initial parameters of the model can be found as Supplementary Table S1 online, while the optimized results are detailed in Supplementary Table S2.

Analysis of experiment results

In this study, MSE, RMSE, MAE, R^2, and VAF were used as evaluation indicators of the prediction model. They can be calculated as follows:

$$MSE = \frac{1}{V}\sum_{t=1}^{V}(\text{ predict}_t - \text{actual}_t)^2 \tag{37}$$

$$RMSE = \sqrt{\frac{1}{V}\sum_{t=1}^{V}(\text{ predict}_t - \text{actual}_t)^2} \tag{38}$$

$$MAE = \frac{1}{V}\sum_{t=1}^{v}(\text{predict}_t - \text{actual}_t) \tag{39}$$

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} \tag{40}$$

$$VAF = \left(1 - \frac{\text{var(error)}}{\text{var(actual)}}\right) \times 100 \tag{41}$$

where V denotes the total number of data in the dataset. $predict_t$ denotes the predicted value of the t_{th} data. $actual_t$ denotes the true value of this data. SS_{res} denotes the residual sum of squares. SS_{tot} denotes the total sum of squares. *var*(*error*) denotes the error variance. *var*(*actual*) denotes the actual data variance.

The MSE, RMSE, MAE, R^2 and VAF of the optimized IBRB-C model are 0.0024, 0.0496, 0.0365, 87.12% and 87.88%, respectively. It can be seen that the optimized model has high accuracy. Experiments validate the selection of attributes of the IBRB-C model and the reference intervals divided by the Kmeans++ algorithm combined with expert knowledge are reasonable and effective.

Ablation experiment

To verify the effectiveness of each module in the IBRB-C model, the same training and testing datasets were utilized to conduct an ablation study. The basic interval BRB model was the baseline, an attribute selection module based on RF, and a reference interval determination module based on expert knowledge and K-means++ algorithm were incrementally incorporated. The traditional BRB model was also considered to prove the superiority of the interval structure.

In the attribute selection module, Random Forest (RF) is used to assess the importance of attributes and select the most valuable ones for the model. This process ensures the validity of input attributes while reducing the number of belief rules. The reference interval determination module generates the boundary points of each interval, leading to a significant improvement in the precision of the model.

The following four models are designed to verify the effectiveness of each module.

Model A: This one is the basic interval BRB model without any added module, using all six available features in the dataset as input attributes.

Model B: The attribute selection module is added to this model. After the attribute selection, Model B has 2 input attributes and 36 belief rules.

Model C: It incorporates the reference interval determination module, which is also the IBRB-C model proposed in this paper.

Model D: This is the traditional BRB model, which has 2 input attributes and 361 belief rules.

The predictive fitting plots for the models are shown in Fig. 5.

Table 6 presents a comparative analysis of Model B and Model D, which correspond to the traditional BRB and the basic interval BRB model, respectively. It is apparently that due to the interval structure, the number of belief rules is reduced by 90% from 361 to 36, and the parameters reduce from 1807 to 216, a decrease of approximately 88%. Crucially, as illustrated in Table 7, Model B shows a considerably enhanced capability for predicting student performance. The advantages of the interval structure are thoroughly demonstrated. Combining the information in Figure 5 and Table 7, the inclusion of each module significantly improves the model's performance. In comparison to Model A, Model B effectively decreases the number of input attributes and enhances accuracy by selecting proper features. The best performance is achieved by Model C through the incorporation of the reference interval determination module. In summary, each module refines the basic model while avoiding redundancy.

Comparative study

To prove the superiority of the proposed IBRB-C model, we compared it with models such as Random Forest(RF), Backpropagation Neural Network(BPNN), K-Nearest Neighbor(KNN), Decision Tree(DT) and Long Short-Term Memory Network(LSTM). Table 8 illustrates the MSE, RMSE, MAE, R^2 and VAF of the models. The results showed that the IBRB-C model exhibits good performance compared to other models: The MSE value is 17.24% and 14.29% lower than RF and BPNN, respectively. The fitting plots of the prediction results of the models are shown in Figs. 6, 7, 8, 9 and 10.

To further validate the robustness of the IBRB-C model, ten cycles of experiments were conducted using the IBRB-C, RF,and BPNN, which were the three best performing models. Figures 11, 12 and 13 illustrate the MSE, RMSE and MAE for each model across the testing cycles. The mean values of MSE, RMSE and MAE for IBRB-C model are 0.0025, 0.0499, and 0.0372, respectively. It is evident that BPNN exhibits greater volatility compared to the other models, while IBRB-C demonstrates relatively stability. On the basis of the overall results, IBRB-C outperforms both RF and BPNN.

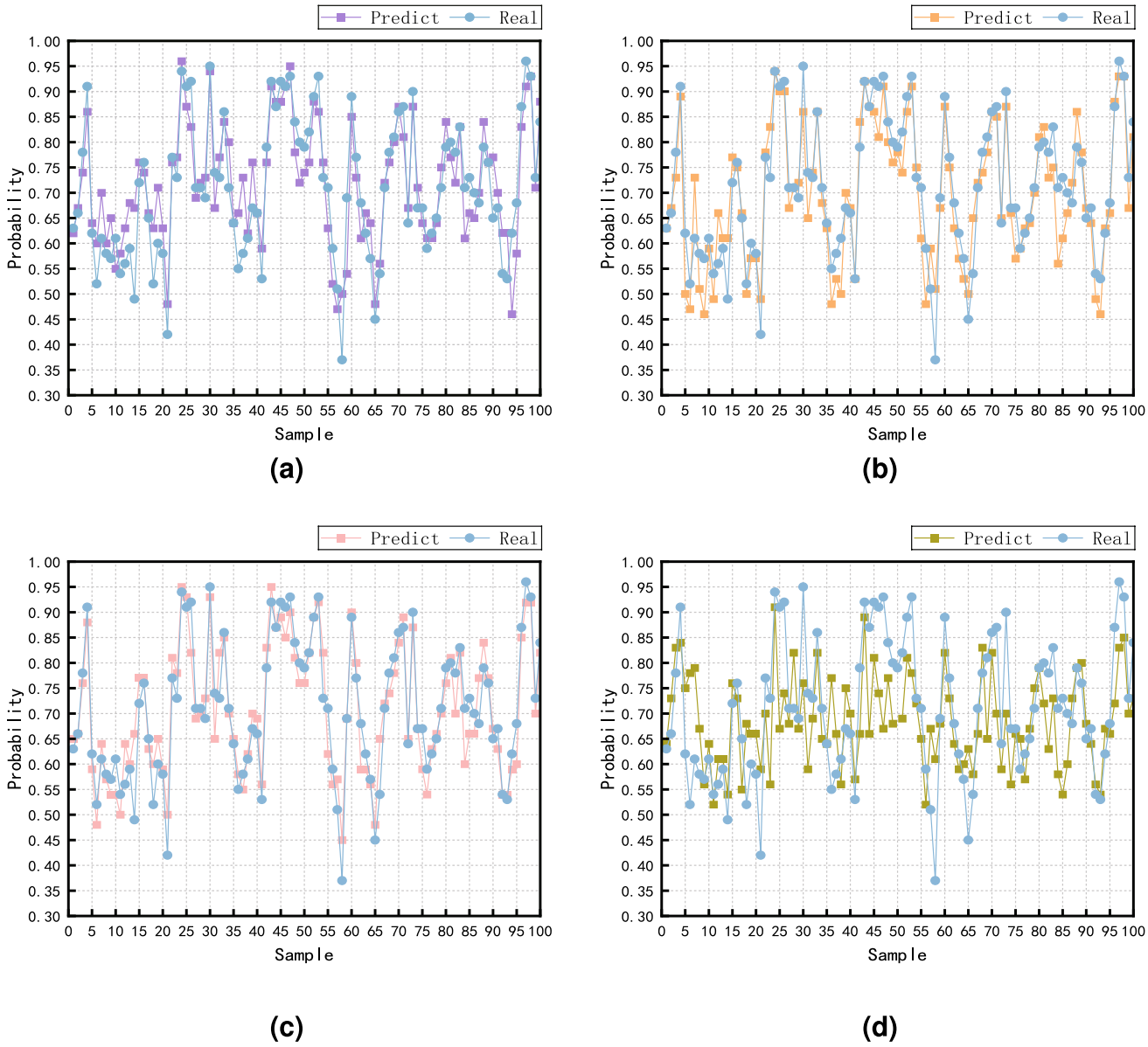


Fig. 5. Predictive fitting plots for four models. (**a**) Model A. (**b**) Model B. (**c**) Model C. (**d**) Model D.

Indicator	Model B	Model D
Input attribute	2	2
Reference interval/point	18	19
Belief rule	36	361
Parameter	216	1807

Table 6. Comparison of Model B and Model D.

IBRB-C model for prediction of students'GPA

To further validate the generalizability of the IBRB-C model, a large-scale dataset was selected to predict the GPA of students. This dataset is publicly accessible via https://www.kaggle.com/datasets/rabieelkharoua/students-performance-dataset (accessed on 13 July 2024), which comprises 2,392 instances. After executing random forest method to analyze attribute importance, Study Time and Absences were chosen as input attributes of the IBRB-C model. The details of attributes are shown in Table 9. According to teaching conventions, five values with segmented significance 0, 1, 2, 3 and 4 were chosen as output levels. 15 reference intervals were defined for

Model	MSE	RMSE	MAE
Model A	0.0037	0.0608	0.0479
Model B	0.0029	0.0539	0.0409
Model C	0.0025	0.0500	0.0372
Model D	0.0143	0.1196	0.0910

Table 7. Comparison of accuracy of four models.

Indicator	IBRB-C	RF	BPNN	KNN	DT	LSTM
MSE	0.0024	0.0029	0.0028	0.0080	0.0035	0.0069
RMSE	0.0490	0.0539	0.0529	0.0894	0.0592	0.0831
MAE	0.0371	0.0410	0.0402	0.0656	0.0445	0.0613
R^2	87.12%	83.93%	84.31%	55.79%	80.78%	61.91%
VAF	87.88%	84.06%	84.91%	55.81%	80.78%	63.19%

Table 8. Comparison of accuracy of IBRB-C with other models.

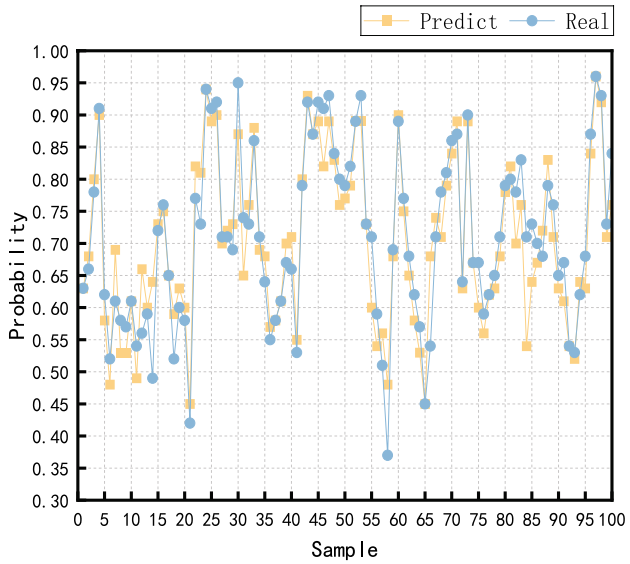


Fig. 6. Predictive fitting plot for RF.

each input attribute, combining the Kmeans++ algorithm with expert knowledge. The initial IBRB-C model was constructed as follows.

$$\begin{aligned} &BeliefRule_c: \\ &if\ x_1 \in [a_1, b_1] \vee x_2 \in [a_2, b_2] \\ &Then\ result\ is\ ((\mathrm{O}_1, \beta_{1,c}), (\mathrm{O}_2, \beta_{2,c}), (\mathrm{O}_3, \beta_{3,c}), (\mathrm{O}_4, \beta_{4,c}), (\mathrm{O}_5, \beta_{5,c})) \\ &with\ rule\ weight\ \omega_c \\ &and\ rule\ reliability\ r_c \\ &\sum_{i=1}^{N} \beta_{i,c} \leq 1, c \in \{1, 2, \ldots, L\} \end{aligned} \tag{42}$$

where x_1 and x_2 represent the Study Time and Absences, respectively. $O_i(i = 1, 2, 3, 4, 5)$ represents the evaluation level of GPA as the prediction result of the model.

In this case study, the same comparative models and evaluation indicators with the subsubsection 4.1.5 are introduced. After conducting ten rounds of experiments, the result and fluctuation range of each evaluation indicator are shown in Table 10 and Fig. 14.

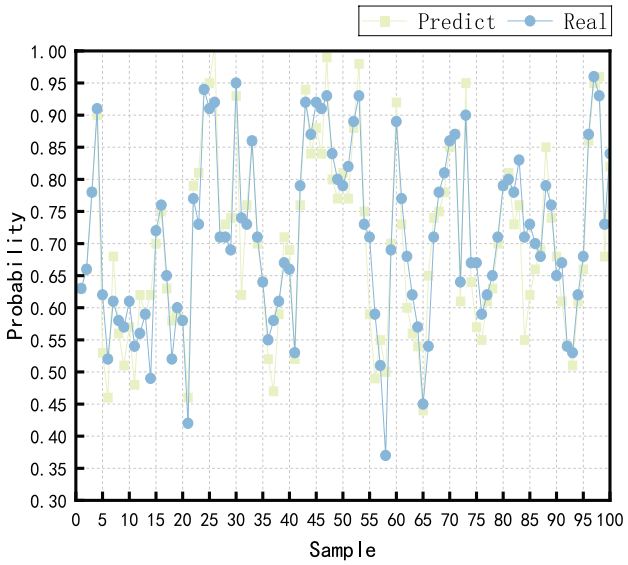


Fig. 7. Predictive fitting plot for BPNN.

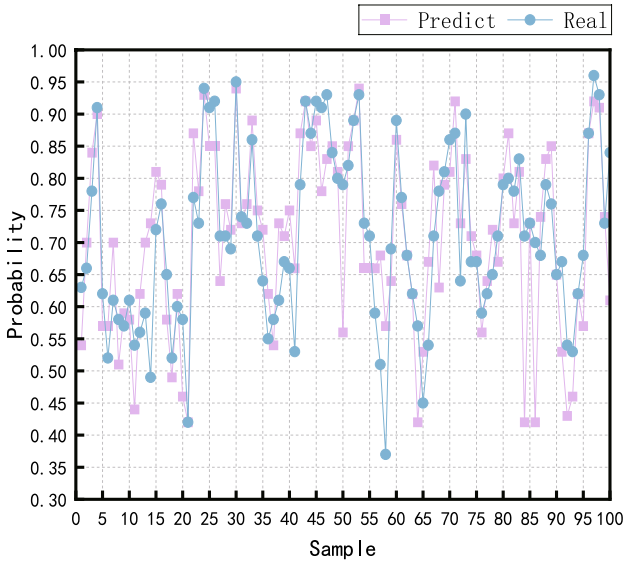


Fig. 8. Predictive fitting plot for KNN.

It is evident that the IBRB-C model still shows the best performance. However, as the sample size of the dataset increases, the advantages of the IBRB-C model have declined. Compared to RF and BPNN, the MSE of IBRB-C decreases by approximately 8.3% and 5.5%, respectively. The possible reason is that with the surge in data volume, the number of belief rules maybe not sufficient, and the initial belief degrees need to be further finely set. Enhancing the effectiveness of the model for large-scale datasets could be a focus of future work. From the fluctuation range of indicators, consistent with the first case study, the IBRB-C model still maintains relatively good robustness compared to RF and BPNN. Furthermore, the prediction results of the LSTM model are significantly superior to its performance in small-sample datasets, while DT exhibited poor performance, with the MSE reaching 0.1462.

In summary, the IBRB-C model exhibits an superior predictive capability across both case studies. In comparison to traditional BRB, IBRB-C not only significantly refines the model structure, reduces the complexity of the rule base, but also greatly improves the accuracy of the results. The five other models introduced in comparative experiments are all commonly used tools in SPP. Nevertheless, due to their reliance on data-driven methodologies, these models are not well-suited for problems involving small sample sizes. Although BPNN and RF show good predictive ability, their results are not interpretable because of the nature of black-box models. The IBRB-C model is constructed through a lucid and transparent process, which is integrated with expert knowledge, enabling educators to easily comprehend how results are derived.

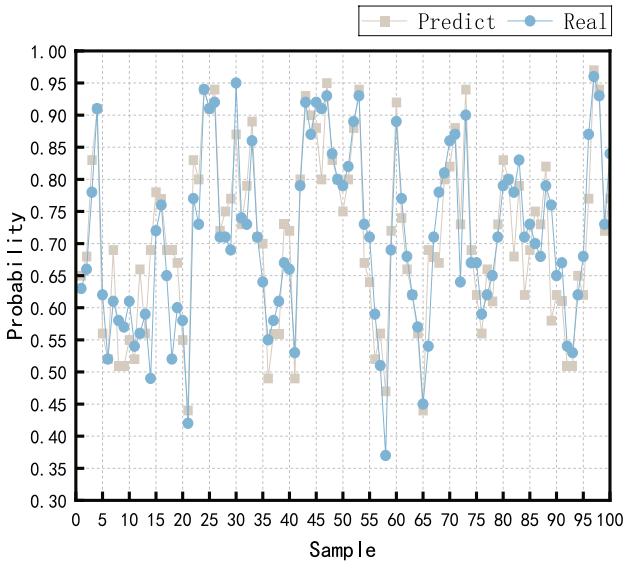


Fig. 9. Predictive fitting plot for DT.

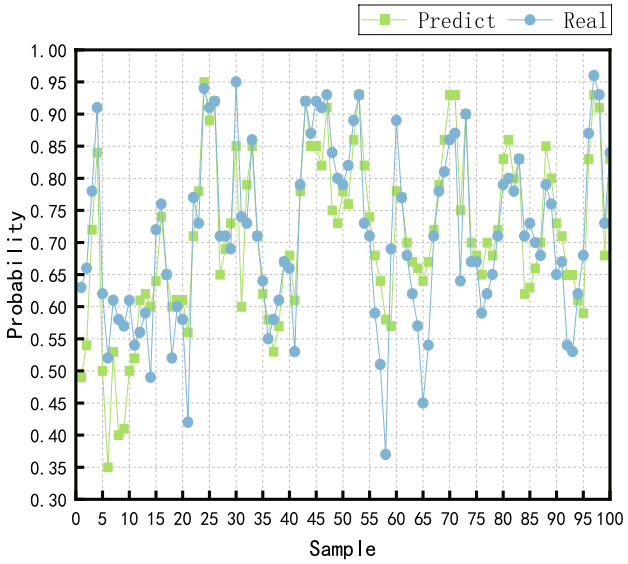


Fig. 10. Predictive fitting plot for LSTM.

Conclusions

In this paper, an IBRB-C student performance prediction model is proposed. The model efficiently manages the combination explosion by employing an RF attribute selection method and segmenting reference intervals. Furthermore, the Kmeans++ algorithm is incorporated to address the constraints of expert experience in practical scenarios, while further research is still needed on how to reasonably select the parameters of the algorithm. By utilizing both expert knowledge and the clustering algorithm, interval boundary points are identified, facilitating a more reasonable construction of reference intervals and improving the model's accuracy. The superiority of the IBRB-C model is validated through comparisons with other models. Owing to the integration of expert knowledge and quantitative information, the model achieves reliable results, particularly on small-scale datasets.

Future work will focus on achieving a balance between the interpretability and precision of the model. Furthermore, we intend to investigate the possible application of the model across other domains. As the number of input attributes increases, how to continue ensuring the prediction accuracy is worth to be studied. We are trying to compare more approaches for establishing reference intervals and develop methods for the dynamic adjustment of interval boundary points, with the objective of augmenting the model's predictive capability.

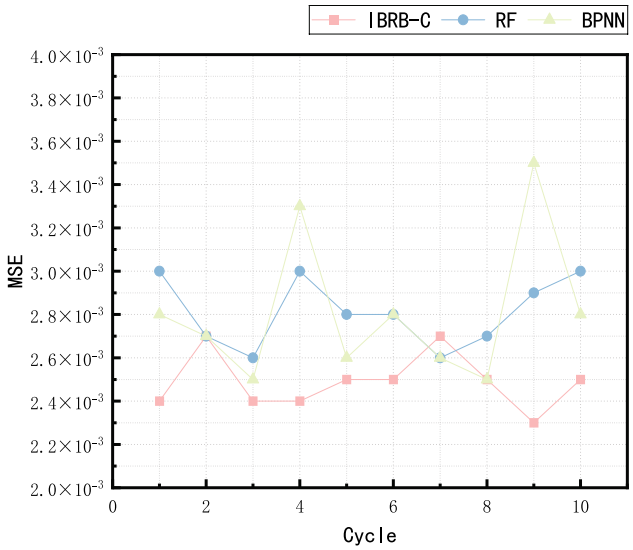


Fig. 11. Comparison of MSE for three models.

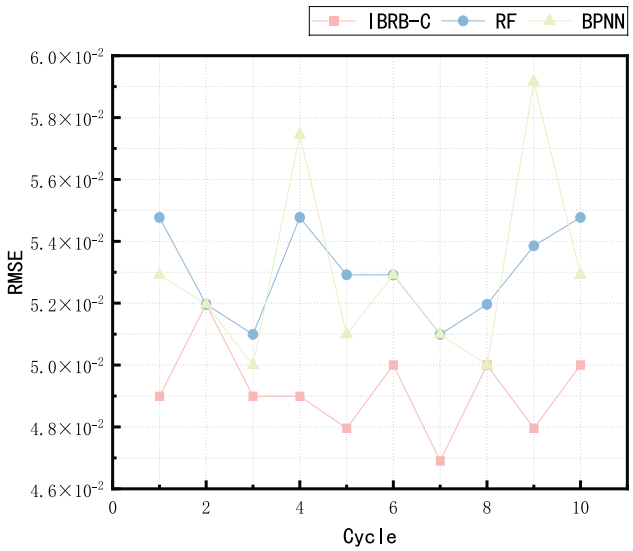


Fig. 12. Comparison of RMSE for three models.

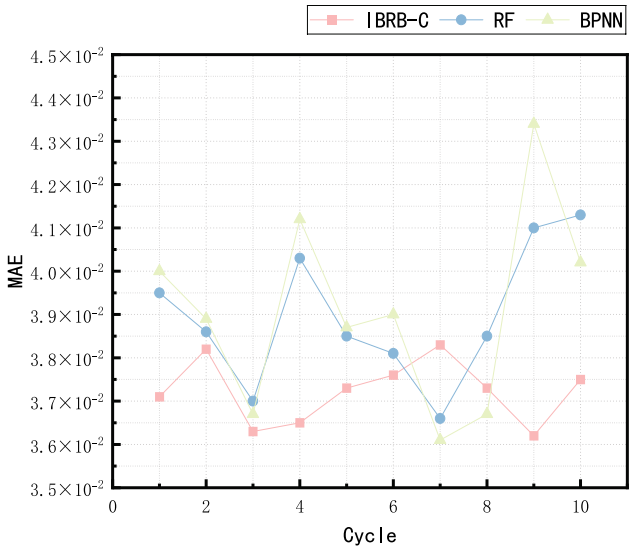


Fig. 13. Comparison of MAE for three models.

Index	Feature	Upper bound	Lower bound	DataType	Type
1	x_1(Study Time)	19.98	0.00	Float	Input
2	x_2(Absences)	29	0	Integer	Input
3	GPA	4.00	0.00	Float	Target

Table 9. The details of input and target attributes.

Indicator	IBRB-C	RF	BPNN	KNN	DT	LSTM
MSE	0.1014 (±0.0024)	0.1106 (±0.0045)	0.1073 (±0.0051)	0.1263 (±0.0062)	0.1462 (±0.0048)	0.1378 (± 0.031))
RMSE	0.3185 (±0.0037)	0.3324 (±0.0067)	0.3273 (±0.0078)	0.3553 (±0.0088)	0.3818 (±0.0063)	0.3689 (± 0.042)
MAE	0.2565 (±0.0034)	0.2673 (±0.0042)	0.2619 (±0.0085)	0.2860 (±0.0076)	0.3068 (±0.0052)	0.2955 (± 0.034)

Table 10. Comparison of MSE, RMSE, and MAE of IBRB-C with other models.

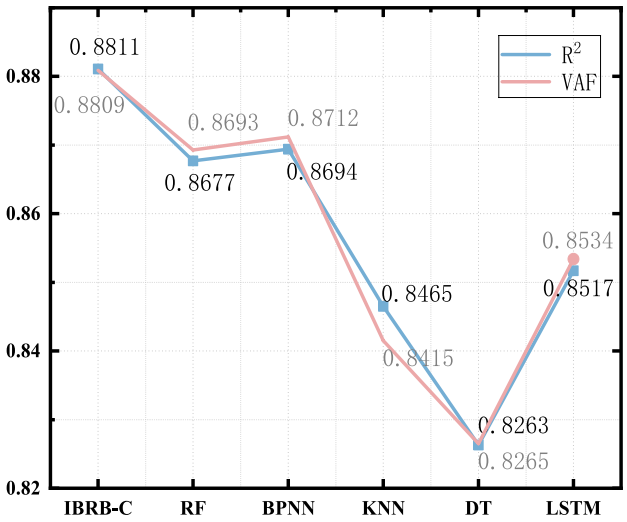


Fig. 14. Percentage indicators of IBRB-C and other models.

Data availability

The datasets analysed during the current study are available in the following links: https://www.kaggle.com/datasets/mukeshmanral/graduates-admission-prediction and https://www.kaggle.com/datasets/rabieelkharoua/students-performance-dataset.

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Competing interests

The authors declare no competing interests.

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